Gradient Descent Solution

Gradient Descent (GD) Algorithm:

[1] Start with initial **Random Weights (W)**

[2] Update the weights by moving along the direction of **negative gradient** of Loss function (w.r.t. W) (**moving step is a hyper parameter, AKA Learning Rate**)

[3] Stop when the gradient is **almost zero** (this is at the minimum value of the function)

Notes:

- Least Squares Linear regression objective function **is a convex function** (even with regularizer). This means it has **ONE Minimum** (good news).

- GD will approximately find this minimum value but at a variable number of steps depending on the used **Learning Rate**